Will 1,100-ft 'God of Chaos' asteroid collide with Earth in 2029?

After extensive tracking and refined calculations, astronomers have claimed that the 1,100 feet wide asteroid, 99942 Apophis, also known as the "God of Chaos," will not collide with Earth as initially predicted in 2004. On April 13, 2029, Apophis will pass by Earth at a distance of approximately 19,794 miles — about the same distance as typical satellites. This close encounter offers scientists an unprecedented opportunity to study Apophis, an ancient asteroid dating back to the early days of the solar system, without any threat of impact.

Predictions and early concerns

When Apophis was first discovered in 2004 by astronomers Roy Tucker, David Tholen, and Fabrizio Bernardi, initial calculations suggested a potential impact with Earth in 2029 and again in 2036, drawing significant public attention. Apophis was even considered one of the most dangerous near-Earth objects (NEOs) at the time. However, radar observations in March 2021 showed that Earth is safe from Apophis for at least a century, dispelling fears of a catastrophic collision.

Apophis flyby in 2029

On April 13, 2029, Apophis will pass closer than the orbit of geosynchronous satellites, creating an extraordinary spectacle. Observers in the Eastern Hemisphere will be able to see the asteroid streak across the sky with the naked eye. NASA plans to redirect the OSIRIS-APophis EXplorer (OSIRIS-APEX) spacecraft from asteroid Bennu to conduct a detailed study of Apophis's surface composition, shape, and rotation speed.

Research goals for the flyby

Scientists believe Apophis has a unique, "bilobed" shape, similar to a peanut, and aim to study this shape to gain insight into its structure and spin mechanics. As Apophis rotates, it's expected to wobble, which researchers hope to observe closely.

This flyby may also trigger minor "asteroid quakes," as Earth's gravitational influence interacts with Apophis's spin and trajectory. OSIRIS-APEX will gather detailed information on the surface's chemical composition, providing valuable data on asteroid formation and the early solar system.

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